

NUTTHAWAT PANYANGNOI CA, (209) 407-9730 | panyangnoi.nutthawat@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Detail-oriented Computer Engineer specializing in Hardware Test Engineering and Embedded Systems. Proven expertise in validating TPU hardware at Google and conducting mission-critical cost analysis for NASA. Proficient in C++, Python, and FPGA development with a strong background in full-stack software and reliability testing.

TECHNICAL SKILLS

Programming: C, C++, Python, JavaScript, MySQL, Verilog, HTML, CSS, Verilog, and FPGA

Libraries: PyTorch, TensorFlow, NumPy, Scikit-learn, Pandas, OpenCV, React, React Native.

Hardware & IT: FPGA Programming, BIOS flashing, RAID, NAS, Oscilloscopes, LTspice, KiCad, Vivado.

Operating Systems: Linux (Kernel customization), Windows, macOS, RTOS.

PROFESSIONAL EXPERIENCES

Hardware Test Engineer (Contractor via Tezerakt LLC) | Google | 10/2025 – Present

- Execute comprehensive reliability protocols (thermal cycling, humidity, burn-in, life testing) on unreleased TPU hardware to validate design integrity.
- Monitor environmental chamber conditions and system-level metrics in real-time to ensure 100% accuracy in simulation without risking hardware damage.
- Analyze performance data to identify hardware vulnerabilities, collaborating with engineering teams to optimize hardware reliability for mass production.

Engineer Student Researcher | NASA L'SPACE / Arizona State University | 01/2024 – 12/2024

- Conducted simulated NASA Discovery Mission cost analysis and timeline scheduling using official tools (MCCET, NICM) to estimate mission resources.
- Co-authored technical research on new space technology high-voltage Lithium-ion battery systems and investigated laser-based charging methods for extended space missions.
- Developed mission concept documentation and feasibility reports, earning NASA Academy and Proposal Writing certifications.

IT Technician | Mountain Cascade, Inc. | 03/2020 – 08/2023

- Maintained 99% uptime for corporate servers through expert hardware repair, system upgrades, BIOS flashing, and RAID configurations.
- Standardized VPN setup and network troubleshooting protocols for engineering and corporate workstations.

Frontend Software Developer | Amber System Technology | 04/2018 – 03/2020

- Engineered scalable full-stack POS web applications, optimizing UI/UX for visual appeal and ease of use for small business clients.
- Led frontend development by collaborating with clients to translate business goals into functional code features.

TECHNICAL PROJECTS

Handheld Computing Device & MIRAI OS | C, Assembly, Linux, KiCad

- Designing a custom handheld computer powered by Raspberry Pi CM4 on a custom-designed Clockwork mainboard.
- Built a lightweight Linux OS including a custom kernel and C programs to interface with MCU via I2C, GPIO, and UART.

RISC-V CPU Pipeline Implementation | Vivado, Verilog, System Verilog

- Designed a 5-stage CPU pipeline for RISC-V architecture, simulating multiplexers, ALUs, data registers, and Finite State Machines (FSM).

Astraeus-1 Satellite Control Board | Embedded C, RTOS, XBee

- Developed embedded C programs for the SJSU Cube3 Satellite Club, enabling communication across multiple microcontrollers and onboard sensors.

EDUCATION & CERTIFICATIONS

B.S. Computer Engineering | San Jose State University | GPA: 3.4/4.0 Graduated: 2025

A.S. Computer Science | Las Positas College | GPA: 3.4/4.0 Graduated: 2022

B.S. Computer Science | King Mongkut's University Thonburi, Thailand| Incomplete due to family relocated

Certifications: Google Foundations of Data Science, NASA L'SPACE Mission Concept Academy, MySQL and Computer Components (LinkedIn).